

Probability

Probability theory is the branch of mathematics that studies the possible outcomes of given events together with the outcomes' relative likelihoods and distributions.

Probability is a way of summarizing the uncertainty of statements or events.

It gives a numerical measure for the degree of certainty (or degree of uncertainty) of the occurrence of an event.

Definitions

- **Experiment:** A process by which an observation or measurement is obtained. Eg tossing a coin or rolling a die.
- **Outcome:** Possible result of a random experiment. Eg a 6 when a die is rolled once or a head when a coin is tossed.
- **Sample space:** Also called the probability space and it is a collection or set of all possible outcomes of a random experiment. Sample space is usually denoted by S, Ω, U
- **Event:** it's a subset of the sample space. Events are usually denoted by upper case letters.

Rules of Probability

1. For any event A in sample space S , $0 \leq P(A) \leq 1$. Consequently if A is empty then $P(A) = P(\emptyset) = 0$ and if $A = S$, $P(A) = P(S) = 1$
2. If A is an event in the sample space S , then (read as 'A complement') is an event in S but outside A .
3. If the sample space S contains n disjoint events $E_1, E_2, \dots, E_n$ then:

$$P(E_1) + P(E_2) + \dots + P(E_n) = \sum_{i=1}^n P(E_i) = 1$$

4. Let A and B be two events such that $A \subseteq B$, then $P(A) \leq P(B)$
5. For any two events A and B $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Examples

1. The Probability that John passes a Maths exam is $4/5$ and that he passes a Chemistry exam is $5/6$. If the probability that he passes both exams is $3/4$, find the probability that he will pass at least one exam.

Solution

Let M be the event that John passes Math exam, and C be the event that John passes Chemistry exam.

$$P(M) = \frac{4}{5}, \quad P(C) = \frac{5}{6}, \quad P(M \cap C) = \frac{3}{4}$$

The probability that he will pass at least one exam is given by:

$$P(M \cup C) = P(M) + P(C) - P(M \cap C)$$

$$P(M \cup C) = \frac{4}{5} + \frac{5}{6} - \frac{3}{4} = \frac{48 + 50 - 45}{60} = \frac{53}{60}$$

2. A fair 6 sided die is rolled twice and the sum of the scores showing up noted. Define A, B and C to be the event that the sum of the scores is greater than 7, a multiple of 3 and a prime number respectively. Show that $P(A \cup B) = P(A) + P(B) - P(AB)$ and also find $P(A \cup C)$, $P(BC)$ and $P(BC')$

Solution

+	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

$$P(A) = \frac{15}{36} = \frac{5}{12} = P(C) \text{ and } P(B) = \frac{12}{36} = \frac{1}{3}$$

$A \cap B$ means the set of all multiples of 3 that are greater than 7. Clearly $P(A \cap B) = \frac{5}{36}$

$A \cup B$ means the set of all values that are multiples of 3 and/or greater than 7. Clearly $P(A \cup B) = \frac{22}{36} = \frac{11}{18} = P(A) + P(B) - P(AB)$

$A \cap C$ means the set of all multiples of 3 that are prime number. Clearly $P(A \cap C) = \frac{2}{36} = \frac{1}{18}$

$$P(A \cup C) = P(A) + P(C) - P(AC) = \frac{5}{12} + \frac{5}{12} - \frac{1}{18} = \frac{7}{9}$$

$B \cap C$ means the set of all greater than 7 that are prime numbers. Clearly $P(BC) = \frac{2}{36} = \frac{1}{18}$

$B \cap C'$ means the set of all greater than 7 that are not prime numbers. Clearly $P(BC') = \frac{11}{36}$

Definition

(Odds of an event) It's the ratio of the probability of an event occurring to that of the event not happening. If A is an event then the odds of A is given by

$$\frac{P(A)}{1 - P(A)}$$

Example

Find $P(A)$ and $P(A^c)$ if the odds of the event is $\frac{5}{4}$

Solution

$$\frac{P(A)}{1 - P(A)} = \frac{5}{4} \Rightarrow 4(1 - P(A)) = 5P(A) \Rightarrow 4 = 9P(A) \Rightarrow P(A) = \frac{4}{9} \text{ and } P(A') = \frac{5}{9}$$

Relationship Between Events

- Mutually exclusive events:** Two events A and B are said to be mutually exclusive if they cannot occur simultaneously. That is if the occurrence of A totally excludes the occurrence of B. Effectively events A and B are said to be mutually exclusive if they disjoint.
- Exhaustive events:** Disjoint events whose union equals the sample space.
- Independent events:** Two events A and B are said to be independent if the occurrence of A does not affect the occurrence of B. If events A and B are independent then $P(AB) = P(A) \times P(B)$

Example

Roll a fair die twice and define A to be the event that the sum of the scores showing up is greater than 7, B be the event that the sum of the scores showing up is a multiple of 3 and C be the event that the sum of the scores showing up is a prime number. Which of the events A, B and C are independent? Are the 3 events jointly independent?

Solution

From the above example, $P(A) = P(C) = \frac{5}{12}$, $P(B) = \frac{1}{3}$, $P(AB) = \frac{5}{36}$ and $P(AC) = P(BC) = \frac{1}{18}$

Since $P(AB) \neq 0$ events A and B are not mutually exclusive. Similarly events A & C and B and C are not mutually exclusive

$$P(A) \times P(B) = \frac{5}{12} \times \frac{1}{3} = \frac{5}{36} = P(AB) \Rightarrow \text{A and B are independent events.}$$

$$P(A) \times P(C) = \frac{5}{12} \times \frac{5}{12} = \frac{25}{144} \neq P(AC) \Rightarrow \text{A and C are dependent events.}$$

$$P(B) \times P(C) = \frac{1}{3} \times \frac{5}{12} = \frac{5}{36} \neq P(BC) \Rightarrow \text{B and C are dependent events.}$$

The 3 events are not jointly independent since pairwise independence is not satisfied.

Theorem 1

If events A and B are independent, then A and B' are also independent.

Proof Decomposing A into two disjoint events AB and AB' . We can write $P(A) = P(AB) + P(AB') \Rightarrow P(AB') = P(A) - P(AB) = P(A) - P(A) \times P(B)$ since events A and B are independent. Thus $P(AB') = P(A)[1 - P(B)] = P(A) \times P(B')$ $\Rightarrow$ A and B' are independent

Theorem 2: If events A and B are independent, then A' and B' are also independent *Proof* Decomposing B' into two disjoint events AB' and $A'B'$. We can write

$$P(B') = P(AB') + P(A'B') \Rightarrow P(A'B') = P(B') - P(AB') = P(B') - P(A) \times P(B')$$

since events A and B' are independent. (From theorem 1 above) Thus

$$P(A'B') = [1 - P(A)]P(B') = P(A') \times P(B') \Rightarrow A' \text{ and } B' \text{ are also independent}$$

Counting Rules Useful in Probability

In some experiments it is helpful to list the elements of the sample space systematically by means of a tree diagram.

In many cases, we shall be able to solve a probability problem by counting the number of points in the sample space without actually listing each element.

Permutations

Permutations refer to an arrangement of objects when the order matters (for example, letters in a word).

The number of permutations of n distinct objects taken r at a time

$${}_nP_r = \frac{n!}{(n-r)!}$$

Example From among ten employees, three are to be selected to travel to three out-of-town plants A, B, and C, one to each plant. Since the plants are located in different cities, the order in which the employees are assigned to the plants is an important consideration. In how many ways can the assignments be made? *Solution;*

Because order is important, the number of possible distinct assignments is ${}_nP_3 = 720$

In other words, there are ten choices for plant A, but then only nine for plant B, and eight for plant C. This gives a total of $10(9)(8)$ ways of assigning employees to the plants.

Combinations

The term combination refers to the arrangement of objects when order does not matter.

For example, choosing 4 books to buy at the store in any order will leave you with the same set of books.

The number of distinct subsets or combinations of size r that can be selected from n distinct objects is given by:

$${}_nC_r = \frac{n!}{r!(n-r)!}$$

Example A university warehouse has received a shipment of 25 printers, of which 10 are laser printers and 15 are inkjet models. If 6 of these 25 are selected at random to be checked by a particular technician, what is the probability that; a) exactly 3 of these selected are laser printers? b) at least 3 inkjet printers?

Solution

First choose 3 of the 15 inkjet and then 3 of the 10 laser printers.

There are ${}_{15}C_3$ and ${}_{10}C_3$ ways to do it, and therefore

$$P(\text{exactly 3 of the 6}) = \frac{{}_{15}C_3 \times {}_{10}C_3}{{}_{25}C_6} = 0.3083$$

$$P(\text{at least 3}) = \frac{{}_{15}C_3 \times {}_{10}C_3}{{}_{25}C_6} + \frac{{}_{15}C_4 \times {}_{10}C_2}{{}_{25}C_6} + \frac{{}_{15}C_5 \times {}_{10}C_1}{{}_{25}C_6} + \frac{{}_{15}C_6 \times {}_{10}C_0}{{}_{25}C_6} = 0.8530$$

Conditional Probability

Conditional probability is the probability of an event occurring given the knowledge that another event has occurred. The conditional probability of event A occurring, given that event B has occurred is denoted by $P(A/B)$ and is read "probability of A given B" and is given by

$$P(A/B) = \frac{P(AB)}{P(B)}$$

Example In a large metropolitan area, the probability of a family owning a colour T.V, a computer or both 0.86, 0.35 and 0.29 respectively. What is the probability that a family chosen at random during a survey will own a colour T.V and/or a computer? Given that the family chosen at random during a survey owns a colour T.V, what is the probability that it will own a computer?

Solution

Let T and C be the event of owning a colour T.V and a computer respectively. Then $P(T \cup C) = P(T) + P(C) - P(TC) = 0.86 + 0.35 - 0.29 = 0.92$

$$P(C/T) = \frac{P(TC)}{P(T)} = \frac{0.29}{0.86} \approx 0.337209$$